



Genetic and epigenetic factors and early life inflammation as predictors of neurodevelopmental outcomes

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ABSTRACT

Among individuals born very preterm, perinatal inflammation, particularly if sustained or recurring, is highly likely to contribute to adverse neurodevelopmental outcomes, including cerebral white matter damage, cerebral palsy, cognitive impairment, attention-deficit/hyperactivity disorder, and autism spectrum disorder. Antecedents and correlates of perinatal inflammation include socioeconomic disadvantage, maternal obesity, maternal infections, fetal growth restriction, neonatal sepsis, necrotizing enterocolitis, and prolonged mechanical ventilation. Genetic factors can modify susceptibility to perinatal inflammation and to neurodevelopmental disorders. Preliminary evidence supports a role of epigenetic markers as potential mediators of the presumed effects of preterm birth and/or its consequences on neurodevelopment later in life. Further study is needed of factors such as sex, psychosocial stressors, and environmental exposures that could modify the relationship of early life inflammation to later neurodevelopmental impairments. Also needed are pharmacological and non-pharmacological interventions to attenuate inflammation towards the goal of improving the neurodevelopment of individuals born very preterm.

1. Introduction

Individuals born very preterm (prior to 32 weeks of gestation) or extremely preterm (prior to 28 weeks of gestation) are at an increased risk for adverse neurocognitive and behavioral outcomes, including cognitive impairment [1], attention-deficit/hyperactivity disorder (ADHD) [1], autism spectrum disorder (ASD) [2], and cerebral palsy (CP). Among individuals born extremely preterm, the risk of severe cerebral palsy, microcephaly, autism spectrum disorder, and severe cognitive impairment is twice as high in males than in females [3].

The developmental origins of health and disease hypothesis posits that early life exposures, including the fetal environment and neonatal inflammation, can have lasting effects on child health, including neurodevelopmental outcomes. Differential CpG methylation of genes in the placenta, cord blood, and neonatal tissue drive molecular mechanisms that can disrupt brain development and lead to neurodevelopmental impairments. Genomic variation also contributes to neurodevelopmental impairments. Increased understanding of the complex relationships among genetics, epigenetic variation, inflammation, and neurodevelopment will inform efforts to improve the neurodevelopment of individuals born very preterm.

As a result of pre-, peri-, and postnatal phenomena, the neonate

delivered extremely preterm is more likely to be exposed to inflammation than infants born closer to term. Among the numerous factors associated with both extreme prematurity and perinatal inflammation are socioeconomic disadvantages [4], intrauterine infection [5–9], fetal growth restriction [10], severe brain hemorrhage [11], prolonged mechanical ventilation [12], necrotizing enterocolitis [13,14], and sepsis [15].

Recent studies on the relationship of perinatal inflammation to neurodevelopmental outcomes have used protein biomarkers as a more direct indicator of inflammation than clinical diagnoses. Based on a meta-analysis of 37 studies, the protein biomarkers most consistently predictive of neurodevelopmental impairments are interleukin-6 (IL-6), interleukin-8 (IL-8), and, to a lesser extent, tumor necrosis factor- α (TNF- α) and interleukin-1 β (IL-1 β) [16].

2. Inflammation disrupts brain development

In preclinical models, treatment with lipopolysaccharide (LPS) or viral mimetics is used to initiate systemic inflammation and neuroinflammation. In experiments performed more than 50 years ago, kittens treated with intraperitoneal injection of LPS developed cerebral white matter damage, but adult cats treated in a similar manner did not [17].

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Similar studies in rodents, rabbits, sheep, dogs, pigs, and non-human primates have elucidated mechanisms linking exposure to LPS or viral mimetics to neuroinflammation and resultant cerebral white matter damage [18–21]. Peripheral immune activation in the early postnatal period can lead to inflammation in the brain, and this relationship is influenced by genetic background, sex, and postnatal age [22]. Pre-clinical models also have been used to study neuroprotective interventions, such as erythropoietin and melatonin, for LPS-induced cerebral white matter injury [23].

In humans, inflammation-related perinatal brain injury is a potential mechanism of brain damage that can be initiated not only by infectious exposures, but also by a wide range of perinatal adversities, such as pre- and post-natal environmental chemicals and treatments given to neonates as a component of neonatal intensive care [4,12,24]. Possibly contributing to the higher risk of neurodevelopmental impairments among males is their heightened inflammatory response to LPS in umbilical cord blood, as compared to female neonates [25].

In preclinical models, sustained or multiple intermittent episodes of inflammation pose a greater threat to the developing brain than do single episodes of inflammation [20]. Increased expression of Toll-like receptors following LPS treatment suggests a mechanism by which an initial exposure to an initiator of inflammation could sensitize the brain to subsequent exposures to additional inflammation initiators [26]. In addition, animal studies have shown that early life inflammation might increase sensitivity to the hazard of inflammation later in life [27].

Human studies also suggest that multiple episodes of inflammation are more injurious to the brain than a single episode. In the Extremely Low Gestational Age Newborn (ELGAN) Study of neonates born before 28 weeks of gestation, infants with both placenta inflammation and neonatal systemic inflammation were more likely to develop brain ultrasound indicators of cerebral white matter damage and developmental impairments at 24 months of age than were infants who had only placenta inflammation or only neonatal systemic inflammation [28]. Similarly, among infants born before 33 weeks of gestation, those with three or more sequential infections were more likely than others to have white matter abnormalities on brain MRI and worse scores on developmental assessments [29].

3. Inflammation and cerebral palsy

Chorioamnionitis (also referred to as chorionitis) was associated with an increased risk of cerebral palsy among low birth weight births in the Collaborative Perinatal Project, which enrolled over 50,000 pregnant women and followed the offspring during childhood [30]. In a meta-analysis of observational studies, clinical chorioamnionitis was associated with an increased risk for cerebral palsy among preterm infants, but the association with histological chorioamnionitis was not statistically significant [odds ratio 1.6; 95% confidence limits 0.9, 2.7] [31].

The ELGAN Study was prompted by evidence of a relationship between perinatal inflammation and disrupted brain development (reviewed above) as well as the observation that levels of several biomarkers of inflammation, such as TNF- α , and Interleukins 1, 6, 8, and 9, are elevated in the first several postnatal days among full-term neonates who subsequently developed cerebral palsy [32]. The strongest association with elevated levels of inflammation biomarkers was with diparesis and the weakest was with hemiparesis. Multiple findings from the ELGAN Study provide support for the concept that perinatal inflammation increases the risk of cerebral palsy among individuals born extremely preterm. First, initiators and biomarkers of infection and inflammation in the placenta were associated with spastic diparesis [33]. Specifically, inflammation of the chorionic plate and inflammation of the chorion/decidua were associated with a more than 2-fold increase in the risk of spastic diparesis, and infants with microorganisms in the placenta, but no histological indications of inflammation, were four times more likely to develop spastic diparesis than infants

whose placenta had neither. Second, maternal cervicovaginal infections during pregnancy were associated with an increased risk of an abnormally low score on a standardized assessment of infant motor skills at 24 months adjusted age [34]. Third, neonatal systemic inflammation was associated with an increased risk of each form of cerebral palsy (quadriparesis, diparesis, and hemiparesis) [35]. Most prominently associated with cerebral palsy was an elevation of IL-8, which also was predictive of cerebral palsy in separate a study of extremely low birth weight infants [36].

4. Inflammation and cognitive disorders

As with individuals born near or at term gestational age, those born very preterm are more likely to have cognitive impairments if they are from families with fewer socioeconomic resources [37]. While indicators of social disadvantage are predictive of systemic inflammation in the neonate delivered extremely preterm, in the ELGAN Study social disadvantage does not appear to confound the relationship between neonatal systemic inflammation and worse cognitive outcomes [4]. Maternal obesity, which is associated with maternal inflammation [38], as well as neonatal inflammation [39], is predictive of worse cognitive and academic abilities among children born extremely preterm [40,41]. Specific microorganisms recovered from placenta, such as anaerobic streptococci and genital mycoplasmas, are associated with both the likelihood of neonatal systemic inflammation [42] and the likelihood of scores indicative of impairment on tests of cognitive abilities and academic achievement [43].

In the ELGAN Study, cognitive impairment is the neurodevelopmental outcome most strongly associated with elevations of inflammation-related proteins in neonatal blood. Sustained neonatal systemic inflammation, defined as elevations of inflammation-related proteins on both postnatal day 7 and postnatal day 14, is associated with an increased risk of severe impairment of early cognitive function at two [44] and ten years of age. This risk increased with an increasing number of inflammation-related protein elevations [45]. While this association could reflect effects of other risk factors for brain damage that are associated with systemic inflammation, such as severe brain hemorrhage, lung injury, necrotizing enterocolitis, and sepsis, elevations of inflammation-related proteins were predictive of severely impaired early cognitive function even when accounting for these neonatal morbidities [46].

An association of sustained neonatal systemic inflammation with cognitive impairment was again found when the ELGAN Study cohort was assessed at ten years of age [47]. Children with elevations of at least 3 inflammation-related proteins had smaller grey matter, brain stems/cerebellar, and total brain volumes than those without the same number of elevations [48]. Elevations of inflammation-related proteins that might serve as “endogenous protectors” against inflammation-related brain disruption [49], such as brain derived neurotrophic factor, were associated with a lower risk of cognitive impairment [47], larger grey matter, white matter, and cerebellum and brainstem volumes [48].

5. Inflammation and ADHD

In the ELGAN Study cohort, lower maternal education, a correlate of systemic inflammation in extremely preterm neonates [4], was associated with a two-fold higher odds of an attention problem identified with the Child Behavior Checklist (CBCL) parent report at 24 months of age. Recovery of mycoplasma from the placenta was associated with a two-fold higher odds of meeting Diagnostic and Statistical Manual of Mental Disorders (DSM) criteria compatible with attention-deficit/hyperactivity problems [50]. Sustained elevations of multiple inflammation-related proteins, including IL-6 and IL-8, were also associated with an increased risk of attention problems identified with the Child Behavior Checklist at 2 years of age. These associations persisted among infants without ultrasound-detectable white matter injury and among

infants without early cognitive impairment [51]. Assessments of the ELGAN Study cohort at 10 years of age included the Child Symptom Inventory-4 (CSI-4), which screens for ADHD. Similar to findings at 24 months, elevated levels, on multiple days in the first postnatal month, of inflammation-related proteins, such as TNF- α and IL-8, were associated with an increased risk of ADHD characteristics as assessed by the child's teacher [99]. Some of this increased risk was modulated by elevations of proteins that might serve as "endogenous protectors", such as IL-6 receptor; regulated on activation, normal T cell expressed and secreted (RANTES); erythropoietin; neutrophilic growth factor; brain-derived neurotrophic factor (BDNF); basic fibroblastic growth factor; insulin-like growth factor-1; platelet growth factor; angiopoietin-1; and angiopoietin-2 [49].

6. Inflammation and epilepsy

By school age, epilepsy had been diagnosed in 7.6% of the ELGAN cohort [52]. Epilepsy was associated with four potential antecedents or correlates of neonatal systemic inflammation: receipt of public health insurance (an indicator of lower socioeconomic status), maternal gestational fever, cerebral white matter disease and severe bronchopulmonary dysplasia. Seizure without epilepsy was associated with indicators of placental infection and inflammation [53].

7. Inflammation and autism spectrum disorder

Preclinical models and epidemiologic studies in humans strongly suggest that maternal immune activation is associated with an increased risk of autism spectrum disorder (ASD) [24]. In the ELGAN Study cohort, severe fetal growth restriction, an antecedent of neonatal systemic inflammation, was associated with an almost 10-fold higher odds for ASD without intellectual disability. Peripartum maternal fever was associated with increased risk of ASD with intellectual disability [54]. Among ELGAN Study participants without intellectual deficit, an elevated concentration of IL-6, on two or more days in the first postnatal month, was associated with a more than two-fold higher odds of ASD. However, this result only occurred when putative "endogenous protectors", such as neurotrophic, angiogenic, or inflammatory proteins, were not elevated. In the presence of a multiple-day elevation of an "endogenous protector", multiple-day elevations of inflammatory proteins (e.g., IL-6, serum amyloid A) were not associated with an increased risk of ASD [55].

8. Genetic polymorphisms, inflammation, and neurodevelopmental outcomes

The most common manifestations of genetic variation are single nucleotide polymorphisms (SNP). SNPs occur in over 90% of human genes and involve a gene locus with two or more possible alleles each having a frequency of at least 1%. Most of the several million SNPs present in each human are shared with a substantial minority of healthy individuals. The cumulative impact of these polymorphisms underlies variation in complex characteristics such as intelligence. Genomic variation is probably the leading cause of mild intellectual impairment [56], and polygenic variation influences the risk of ASD and ADHD [57,58]. However, the contribution of genetic factors to neurodevelopmental outcomes after very preterm birth has not been well studied.

Some genetic polymorphisms influence inflammatory and anti-inflammatory responses. For example, IL-17A (rs2275913) and IL-17F (rs763780) gene polymorphisms are associated with inflammatory disorders [59]. In a sample consisting of 360 children with ADHD, 21 affected siblings and 17 unaffected siblings, with a mean birthweight of 3389 and mean gestational age 40 weeks, SNPs within genes related to inflammation moderated the relationship between birthweight and ADHD symptom severity in children [60]. Studies of relationships between SNPs and necrotizing enterocolitis and sepsis have been limited

by sample size [61,62], but suggest the need for further study.

Two studies have identified that a polymorphism in the gene for methylenetetrahydrofolate reductase was associated with a higher risk of severe neonatal brain hemorrhage [63,64]. In a case-control study nested within a randomized trial that enrolled mothers with preterm labor before 32 weeks of gestation, a SNP within the gene for IL6R was associated with a more than 3-fold increase in the risk of psychomotor delay, and three SNPs in the IL6R gene were associated with mental developmental delay [65]. A SNP in the gene for toll-like receptor-4 was associated with a 5-fold increase in the risk of cerebral palsy. Studies of very preterm infants have identified associations between cerebral palsy and SNPs in the genes C-C motif chemokine ligand 18 (CCL18) [66], TNF α , and IL1 β [67]. A SNP in the mannose binding lectin gene was associated with worse neurodevelopmental outcome during infancy [68]. In an example of gene-environment interaction, higher cumulative exposure to morphine during the neonatal hospitalization has been associated with behavioral problems at 18 months corrected age, but only among individuals with genetic variants that affect accumulation of morphine in the brain [69].

9. Epigenetic marks and inflammation

Epigenetic marks do not alter the nucleotide sequence of DNA but influence gene expression through microRNAs, histone modification, and DNA methylation [70]. MicroRNAs are RNAs of about 22 nucleotides in length that can control translation and degrade proteins. Histone modifications cause activation and deactivation of transcription, and include acetylation, methylation, and phosphorylation. Most often DNA methylation involves the addition of methyl groups to cytosine at DNA promoter regions where a cytosine nucleotide is followed by a guanine nucleotide, and is known as CpG methylation. Generally, hyper-methylation decreases gene expression, while hypo-methylation increases gene expression. Epigenetic modifications occur normally during development, can occur in response to environmental exposures, and are associated with a plethora of health outcomes.

Evidence that epigenetic changes could influence neuroinflammation includes the findings that: 1) In the brains of patients with late onset Alzheimer disease, genes encoding IL-6 and IL-1 β were hypomethylated, and levels of mRNA for the corresponding proteins were higher than those of controls [71]. 2) In a rat model of temporal lobe epilepsy, expression of miR-146a is increased in reactive astrocytes, which are cells that play a central role in neuroinflammation [72]. 3) In mice, infection with Japanese encephalitis virus increases miR-19b-3p and miR-15b, which, in turn, are associated with increased inflammation [73,74]. Indirect evidence that inflammation could influence epigenetic marks comes from an ELGAN study report indicating that recovery of *Lactobacillus* from placenta, which has been associated with decreased neonatal systemic inflammation and a decreased risk of cognitive impairment among children born extremely preterm [42,43], is correlated with variation in placental DNA CpG methylation of inflammation-related genes [75].

10. Epigenetic marks and neurodevelopmental outcomes

Preclinical models provide strong support for epigenetics as a link between early life exposures and neurodevelopmental outcomes [76]. Among very preterm infants, increased methylation of the serotonin transporter gene (SLC6A4) in neonatal blood obtained at hospital discharge was associated with exposure to painful procedures during hospitalization, greater emotional negativity identified with the face-to-face still face paradigm, and total child behavior problems on the CBCL at 7 years of age [77,78]. Increased DNA methylation of the glucocorticoid receptor gene in the placenta and placental microRNA expression profiles have been associated with neonatal neurobehavior, such as NICU Network Neurobehavioral Scale (NNNS) attention and quality of movement scores [79,80]. A meta-analysis of studies not

focused on very preterm birth provides preliminary support for an association between epigenetic alterations and both ADHD and ASD [81]. In a sample of term-born children studied at 4 years of age, increased methylation of CpG sites within a homologue of *Drosophila* hairy and enhancer of split gene1 (*HES1*), a gene expressed in neural stem cells during brain development, was associated with a higher intelligence quotient. In a replicate sample of children studied at 7 years of age, greater methylation within *HES1* was associated with higher executive memory function [82]. In the ELGAN cohort, differential methylation in the placenta at birth related to cognitive outcomes at age 10. Sixteen unique genes were found to contain CpG hypermethylated probes that were associated with severe cognitive impairment at age 10. Many of these genes were tied to neurodevelopmental outcomes, including G protein subunit alpha o1 (*GNAO1*) and ras homologue family member F (*RHOF*), with increasing odds of moderate or severe cognitive impairment being predicted by probe site increases in methylation [83]. In a separate study involving the ELGAN cohort, CpG methylation levels at 41 loci, representing 10 hypothalamic-pituitary-adrenal axis genes, were associated with moderate/severe cognitive impairment at 10 years of age. Methylation levels within the gene for brain-derived neurotrophic factor, which plays critical roles in brain development and function, also were associated with cognitive outcomes in the ELGAN cohort [84].

In summary, emerging evidence suggests that prenatal and postnatal environmental exposures can influence epigenetic markers in the infant and epigenetic markers early in life are predictive of neurodevelopmental outcomes years later. In this context the environment can be viewed broadly to include the psychosocial, chemical, and physical environment to which the mother is exposed; the intrauterine environment, which is influenced by maternal health and chemicals to which she is exposed; the postnatal environment, which in the case of very preterm infants includes chemicals to which the neonate is exposed during neonatal intensive care [85]; systemic inflammatory responses initiated by complications of very preterm birth (e.g., necrotizing enterocolitis) [13]; and developmentally supportive care. If further research confirms the importance of epigenetic markers as a mechanism for the “early life” origins of chronic illness [86–91], interventions to interrupt this mechanism hold potential for improving the health and development of individuals born very preterm.

11. Implications for clinical care

- Evidence that placental and neonatal epigenetic marks are predictive of neurodevelopmental outcomes suggests a mechanism for, and imparts biological plausibility to, transgenerational effects of the early caregiving environment, potentially including that provided by fathers [92–95]. Transgenerational effects of parental stress implies that efforts to prevent and ameliorate psychosocial stressors could have effects on multiple generations, amplifying the public health benefits.
- While the finding of consistent associations between early life inflammation and neurodevelopmental impairments suggests a potential target for therapies to improve neurodevelopmental outcomes for very preterm neonates, specific intervention strategies have not been identified. Candidate molecules to serve as immune modulators include erythropoietin [96], melatonin [23,97], and N-acetylcysteine [98].
- Even more important are strategies to prevent perinatal initiators of inflammation, such as prolonged mechanical ventilation, necrotizing enterocolitis, and bacteremia.

12. Implications for research

- Given compelling evidence that perinatal inflammation disrupts brain development, investigations in preclinical models of neuroinflammation offer a promising avenue for discovery of effective

neuroprotective interventions to decrease the burden of neurodevelopmental impairment among individuals born very preterm.

- In preclinical models, early life inflammation can sensitize an animal to a subsequent inflammation initiator, which could explain why postnatal episodes of inflammation are more hazardous if preceded by a prenatal inflammation initiator. If this “multiple hit” model of pathogenesis is found to apply to human neonates, postnatal interventions targeting the molecular processes that underlie sensitization could improve neurodevelopmental outcomes.
- Specific biomarkers in neonatal blood are predictive of a range of neurodevelopmental impairments. Incorporating measurements of such biomarkers into clinical trials could enhance the information derived from such trials.

Declaration of competing interest

The authors have no financial or personal relationships with other people or organizations that could bias the content of this article.

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